

● EASY

● ALGEBRA

● ANSWER KEY

Linear equations in one variable

03

10 answers · Rationale-rich · Teach from doc

Q1**D****D.** 3

Choice D is correct. Subtracting $16x$ from both sides of the given equation yields $24 = 8x$. Dividing both sides of this equation by 8 yields $3 = x$. Therefore, the solution to the given equation is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2**A****A.** -1

Choice A is correct. Subtracting 8 from both sides of the given equation yields $p + 3 = 2$. Subtracting 3 from both sides of this equation yields $p = -1$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**B****B. Exactly 1**

Choice B is correct. Subtracting 4 from each side of the given equation yields $6 = 2x$, or $x = 3$, so the equation has a unique solution of $x = 3$.

Choice A is incorrect. Since 3 is a value of x that satisfies the given equation, the equation has at least 1 solution. Choice C is incorrect. Linear equations can have 0, 1, or infinitely many solutions; no linear equation has exactly 3 solutions. Choice D is incorrect. If a linear equation has infinitely many solutions, it can be reduced to $0 = 0$. This equation reduces to $x = 3$, so there is only 1 solution.

Q4**0.2**

Accept also: 1/5

The correct answer is .2 . Subtracting 2.6 from each side of the given equation yields $x = 0.2$. Therefore, the value of x that's the solution to the given equation is 0.2. Note that .2 and 1/5 are examples of ways to enter a correct answer.

Q5**B****B. 40**

Choice B is correct. It's given that the shop's inventory starts with 4,500 paper cups and that the manager estimates that 70 of these paper cups are used each day. Let x represent the number of days in which the estimated supply of paper cups will reach 1,700. The equation $4,500 - 70x = 1,700$ represents this situation. Subtracting 4,500 from both sides of this equation yields $-70x = -2,800$. Dividing both sides of this equation by -70 yields $x = 40$. Therefore, based on this estimate, the supply of paper cups will reach 1,700 in 40 days.

Choice A is incorrect. After 20 days, the estimated supply of paper cups would be $4,500 - 70(20)$, or 3,100 cups, not 1,700 cups.

Choice C is incorrect. After 60 days, the estimated supply of paper cups would be $4,500 - 70(60)$, or 300 cups, not 1,700 cups.

Choice D is incorrect. After 80 days, the estimated supply of paper cups would be $4,500 - 70(80)$, or -1,100 cups, which isn't possible.

Q6**B****B. 8**

Choice B is correct. Dividing each side of the given equation by 3 yields $x - 9 = 8$. Therefore, the value of $x - 9$ is 8.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $3x - 27$, not $x - 9$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**A****A. 7**

Choice A is correct. Applying the distributive property on the left-hand side of the given equation yields $3x + 5x + 20 = 76$, or $8x + 20 = 76$. Subtracting 20 from each side of this equation yields $8x = 56$. Dividing each side of this equation by 8 yields $x = 7$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the solution to the equation $x + 4 = 76$, not $3x + 5x + 4 = 76$.

Q8**21**

The correct answer is 21. It's given that the equation $3x + 21 = 3x + k$ has infinitely many solutions. If an equation in one variable has infinitely many solutions, then the equation is true for any value of the variable. Subtracting $3x$ from both sides of the given equation yields $k = 21$. Since this equation must be true for any value of x , the value of k is 21.

Q9**A****A. 11**

Choice A is correct. Dividing both sides of the given equation by 8 yields $x = 11$. Therefore, 11 is the solution to the given equation.

Choice B is incorrect. This is the solution to the equation $x + 8 = 88$.

Choice C is incorrect. This is the solution to the equation $x - 8 = 88$.

Choice D is incorrect. This is the solution to the equation $\frac{x}{8} = 88$.

Q10

C

C. $16p + 37 = 165$

Choice C is correct. It's given that John made a \$ 16 payment each month for p months. The total amount of these payments can be represented by the expression $16p$. The down payment can be added to that amount to find the total amount John paid, yielding the expression $16p + 37$. It's given that John paid a total of \$ 165. Therefore, the expression for the total amount John paid can be set equal to that amount, yielding the equation $16p + 37 = 165$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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